Significance Sorting in Government Intervention Research: Evidence from Top Finance Journals*

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ABSTRACT

We document a significance-sorting pattern in the three leading finance journals—the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics*—using a novel sample of government intervention research spanning 2000–2024. We identify 200 in-scope papers and code the direction of each paper's primary empirical finding. Papers that report any directional finding are published at a rate of approximately 88%, compared with only 48% for papers that report null results—a gap of roughly 40 percentage points that is robust across specifications and sub-periods. Sorting operates entirely on whether a paper detects an effect, not on the sign of that effect: positive-finding and negative-finding papers reach top journals at virtually identical rates (88.2% and 87.7%). The pattern is concentrated in the *Review of Financial Studies* and the *Journal of Financial Economics*. An analysis of American Finance Association conference papers, which precede journal submission, reveals no comparable gap, consistent with the premium arising at the editorial review stage rather than from differential submission by authors. We interpret our findings

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as a cross-sectional association and are explicit that our design cannot distinguish editorial selection from author self-selection in submission.

JEL classification: G18, G28, C80, B40.

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1. Introduction

Academic research on government intervention in financial markets occupies a privileged position in public policy. The Federal Reserve Board, the Securities and Exchange Commission, and the Basel Committee on Banking Supervision routinely incorporate peer-reviewed finance research into regulatory impact assessments, rule-makings, and policy speeches. The Dodd-Frank Wall Street Reform and Consumer Protection Act alone generated an estimated 400 new regulatory mandates, many of which were designed in explicit reference to academic evidence on the effects of capital requirements, disclosure rules, and consumer financial protection. The credibility of this evidence base is therefore a first-order policy question: if the academic record systematically misrepresents which interventions have measurable effects—and which do not—regulators face a distorted picture that can generate welfare-reducing policy choices.

Publication bias offers a mechanism by which such distortion arises without any deliberate misconduct. When researchers are reluctant to write up null results, when journals are more likely to accept papers with statistically significant findings, or when both effects operate simultaneously, the published literature understates the true prevalence of null or inconclusive evidence (Franco et al., 2014). In a domain such as government intervention research, where policy decisions depend on whether a regulation has a measurable causal effect, suppression of null results generates the false impression that most interventions work—or, more precisely, that most interventions have discernible effects in one direction or the other.

This paper studies publication bias in a domain the existing literature has not examined: the finance literature's treatment of government intervention. Prior work on publication bias in finance has focused on the cross-section of asset pricing anomalies, documenting that factor-zoo proliferation reflects selection for high t-statistics (Harvey et al., 2016; Hou et al., 2020; Jensen et al., 2023). A parallel economics literature (Brodeur et al., 2016; Andrews and Kasy, 2019) infers bias from the shape of published p-value distributions, finding excess mass just below conventional significance thresholds. Neither strand addresses the policy-intervention literature, nor do they distinguish two conceptually distinct forms of bias with different welfare implications. *Significance bias*—directional findings over-published relative to null findings—makes every intervention appear detectable. *Directional bias*—pro-intervention findings over-published relative to anti-intervention findings—additionally skews the regulatory record toward a particular conclusion about the sign of policy effects. We design our study to test for both.

Our research design combines two complementary samples. The *published paper* sample consists of all in-scope articles published in the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics* from 2000 to 2024. The *working paper* sample is drawn from NBER working papers in five programs: Corporate Finance (CF), Asset Pricing (AP), Monetary Economics (ME), Economic Fluctuations and Growth (EFG), and Public Economics (PE), covering papers from 2000 to 2024. A key methodological advantage of this two-sample design is that we observe actual pre-publication findings rather than merely the shape of the published distribution: the gap between pre- and post-publication finding rates is directly observable rather than inferred from p-value bunching, though the cross-sectional design cannot rule out that it reflects differences between the two populations rather than the editorial filter alone. Together, these samples allow us to compare the full distribution of findings across publication states—not merely the subset that appears in top journals.

We define a paper as in-scope if its primary research question is the causal effect of a specific government law, regulation, or public policy on financial markets, firms, or investors. Identifying in-scope papers in a corpus of more than 33,000 documents would be infeasible by hand. We therefore develop a two-pass automated pipeline: a keyword filter that screens for government intervention terminology, followed by a large language model (LLM) classifier that reads each flagged abstract and returns a binary in-scope decision with a structured justification. For each in-scope paper we then code the *direction* of the primary finding as positive (+1, the intervention was beneficial), negative (−1, harmful), or null/mixed (0).

Our primary finding is a pronounced *significance-sorting* pattern in the published record: papers with *any* directional finding appear in top finance journals at substantially higher rates than null-finding papers. A binary directional indicator (directional vs. null, sidestepping normative positive/negative coding) carries a probit coefficient of 1.231^{***} (s.e. 0.204, $p < 0.001$), corresponding to a 40 percentage-point gap in direction-specific publication rates. Within directional papers, positive-finding papers are published at a rate of 88.2% and negative-finding papers at 87.7%—both substantially exceeding the 48% rate for null papers. A χ^2 test of homogeneity strongly rejects equality across the three direction categories ($\chi^2 = 38.456$, $p < 0.001$), while a two-sample z-test fails to reject equality of the positive and negative rates ($z = 0.09$, $p = 0.93$). We cannot detect directional asymmetry—both types of directional finding are associated with similar publication rates—though at our sample size the test has limited power to detect moderate asymmetries (minimum detectable difference 0.851 coefficient units at 80% power).

Probit models confirm this pattern. The coefficient on an indicator for a positive finding is 1.247 ($p < 0.001$), implying a marginal publication probability approximately 41 percentage points above that of a null-finding paper. The coefficient on a negative finding indicator is 1.219 ($p < 0.001$), implying a nearly identical 40 percentage-point premium.[1] The two direction coefficients are statistically indistinguishable ($p = 0.93$). We note, however, that this test has limited power at our sample size: given the observed standard errors, the minimum detectable difference between $\hat{\beta}_1$ and $\hat{\beta}_2$ at 80% power is 0.851 coefficient units (SE of the difference = 0.304). We can therefore rule out large asymmetries, but cannot exclude directional differences smaller than this threshold. Within this constraint, the evidence is consistent with a significance-sorting pattern—not a directional one.

[1] Marginal effects for Cols. (2)–(6) are evaluated at the bivariate-spec null-paper baseline ($\hat{\alpha} = -0.060$, implied publication probability $\Phi(-0.060) \approx 47.6\%$). The effect for positive-finding papers is $\Phi(-0.060 + 1.247) - \Phi(-0.060) = 88.2\% - 47.6\% \approx 41$ pp; for negative-finding papers, $\Phi(-0.060 + 1.219) - \Phi(-0.060) = 87.7\% - 47.6\% \approx 40$ pp. For Col. (1), the binary-spec baseline is $\hat{\alpha} = -0.060$ ($\Phi(-0.060) \approx 47.6\%$), giving a directional gap of $\Phi(-0.060 + 1.231) - \Phi(-0.060) \approx 88.0\% - 47.6\% \approx 40$ pp.

The quasi-experimental identification control in Column (6) confirms that identification quality is strongly associated with publication: the QuasiExp dummy carries a coefficient of 0.734^{***} ($p < 0.001$), consistent with clean-identification papers having a substantial structural advantage in the review process—an advantage that is separate from and additive to the direction premium. Crucially, however, the direction coefficients are essentially unchanged ($\hat{\beta}_1 = 1.228^{***}$, $\hat{\beta}_2 = 1.228^{***}$), confirming that the significance premium is not an artifact of null-finding papers relying on weaker identification strategies.

The structure of the significance-sorting pattern varies across journals. The direction–publication association is large and highly significant in both the *Review of Financial Studies* ($\hat{\beta}_{\text{pos}} = 1.287^{***}$, $\hat{\beta}_{\text{neg}} = 1.459^{***}$) and the *Journal of Financial Economics* ($\hat{\beta}_{\text{pos}} = 1.448^{***}$, $\hat{\beta}_{\text{neg}} = 1.232^{***}$). The *Journal of Finance* coefficients are positive, statistically significant, and economically meaningful ($\hat{\beta}_{\text{pos}} = 1.150^{***}$, $\hat{\beta}_{\text{neg}} = 0.900^{**}$, both $p < 0.05$), but *smaller in magnitude* than those in the *Review of Financial Studies* and *Journal of Financial Economics*. Point estimates are consistent with a smaller association at the *Journal of Finance*, though cross-journal differences are not statistically distinguishable at available sample sizes (see Section 5).

The exploratory sub-period analysis reveals time variation in the structure of the pattern. During the financial crisis and post-crisis period (2008–2015), the negative-finding premium ($\hat{\beta} = 1.307^{***}$) exceeded the positive-finding premium ($\hat{\beta} = 1.148^{**}$), though the gap is smaller than before, consistent with but not causally identifying elevated publication rates for research on the unintended costs of the post-crisis regulatory expansion. In the post-Dodd-Frank period (2016–2024), the two premiums converge to 0.995 and 1.022, respectively, as the publication pattern settled into a symmetric significance-sorting regime. These sub-period estimates are based on small sub-samples ($N = 28$, 69, 103) and should be interpreted as descriptive, not confirmatory.

This paper contributes to three strands of the literature. It provides systematic evidence on publication bias in the government-intervention finance literature and identifies the *form* it takes: significance bias—sorting on any directional finding relative to null results—rather than directional bias based on the sign of findings. This distinction matters for policy. Significance bias inflates the apparent measurability of policy effects, creating the impression that most interventions have detectable consequences when the true share of null results is substantially higher. Because neither pro-intervention nor anti-intervention findings carry a differential premium, the regulatory record is not systematically skewed toward a particular policy conclusion; the distortion is about *detectability* rather than *direction*.

The paper also contributes methodologically to the literature on identifying publication bias. Unlike approaches that infer bias from the internal shape of published p-value distributions, our NBER working paper baseline allows us to observe actual pre-publication findings and measure the sorting gap in the cross-section. The AFA conference test provides a useful bounding exercise: at the pre-journal presentation stage directional papers carry no detectable association with eventual publication ($\hat{\beta} = 0.077$, $p = 0.80$); the association is large only in the journal publication comparison ($\hat{\beta} = 1.231^{***}$), consistent with the gap arising at or after the journal review stage rather than earlier in the research process—though supply-side submission decisions at that stage cannot be ruled out.

Finally, the paper introduces a scalable LLM-assisted pipeline for classifying and direction-coding large economics paper corpora. Throughout, we are explicit that our design identifies a cross-sectional association between finding direction and publication status; it cannot distinguish editorial selection from author self-selection in submission or specification search.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes data construction and the classification pipeline. Section 4 presents the empirical strategy. Section 5 reports main results. Section 6 presents robustness checks. Section 7 discusses economic magnitudes, mechanisms, and limitations. Section 8 concludes.

2. Related Literature

2.1 Publication Bias in Asset Pricing

The closest prior work on publication bias in finance examines the cross-section of expected returns. Harvey et al. (2016) survey hundreds of return predictors published in top finance journals and argue that the conventional $t > 2$ threshold is inadequate given the scale of multiple testing in the factor zoo. Hou et al. (2020) attempt to replicate 452 published anomalies and find that 65% fail under standard portfolio construction choices, consistent with upward-biased published effect sizes. Jensen et al. (2023) document that most factors replicate out-of-sample but that the factor space is highly redundant, implying substantial duplication in published claims. This strand focuses entirely on the asset pricing domain and diagnoses bias through t-statistic distributions or replication rates. It does not study policy-intervention research, and it cannot distinguish significance bias from directional bias because factor-return sign is not normatively coded.

2.2 Publication Bias in Economics

The economics literature on publication bias draws on a long tradition in statistics and medicine. Brodeur et al. (2016) analyze 50,000 hypothesis tests reported in the *American Economic Review*, the *Journal of Political Economy*, and the *Quarterly Journal of Economics*, finding excess mass just above the 5% and 10% significance thresholds consistent with selective reporting. Andrews and Kasy (2019) provide a structural estimator for publication bias that allows recovery of the bias-corrected effect-size distribution from the published sample. Ioannidis et al. (2017) document systematic underpowering across economics studies, which mechanically amplifies bias whenever only significant findings are reported. Christensen and Miguel (2018) survey the broader reproducibility literature and document that publication bias, p-hacking, and HARKing (hypothesizing after results are known) are pervasive in empirical economics. These approaches infer bias from the internal shape of the published distribution—they do not observe pre-publication findings and therefore cannot attribute the sorting gap to editorial selection specifically. Our paper contributes domain-specific evidence for the finance-regulation literature using a pre-publication baseline that identifies the editorial stage directly.

2.3 Government Intervention in Financial Markets

Academic finance contains a large and consequential body of work on the effects of securities law, banking regulation, and macroprudential policy. Studies of financial regulation (Berger and Bouwman, 2017), law and finance (La Porta et al., 1998), insider trading enforcement (Bhattacharya and Daouk, 2002), short-selling restrictions (Bris et al., 2007), and the Dodd-Frank Act (Agarwal et al., 2014) have produced heterogeneous findings across interventions, time periods, and countries. Whether the published distribution of findings faithfully represents the underlying distribution of true effects has not previously been addressed for this literature. We fill this gap.

2.4 LLMs in Economics Research

Recent work demonstrates that large language models can accurately classify economic texts, extract structured data from unstructured documents, and replicate tasks that previously required extensive human coding (Ash and Hansen, 2023; Korinek, 2023). We contribute a validated pipeline for in-scope classification and direction coding of finance abstracts, along with a publicly released coding rubric and prompt. Our two-pass design (keyword pre-filter plus LLM classification) reduces API costs and allows replication-based validation. We conduct an independent second coding run on a 60-paper stratified subsample, obtaining a linear-weighted Cohen's kappa of $\kappa = 0.674$, consistent with "substantial agreement" by the Landis and Koch (1977) classification, and addressing concerns about LLM reliability in structured classification tasks (Brodeur et al., 2023).

3. Data and Sample Construction

3.1 Published Papers

The published-paper sample consists of all articles appearing in the *Journal of Finance*, the *Review of Financial Studies*, and the *Journal of Financial Economics* from 2000 to 2024. These three journals constitute the traditional top tier of academic finance and account for the majority of highly cited finance research on government intervention. We collect article metadata—title, authors, year, volume, issue, DOI, and abstract—for all articles in each journal over this period.

Journal articles are identified and retrieved as follows. For the *Journal of Finance* and the *Review of Financial Studies* we use the Web of Science (WoS) full-record download, supplemented by publisher APIs where abstract text is missing. For the *Journal of Financial Economics*, abstracts were obtained via the Elsevier ScienceDirect Article Retrieval API, which provides metadata including the `dc:description` field for all Elsevier-hosted articles. Table 1 summarizes the journal sources.

Table 1. Journal Sample

Journal	Abbrev.	ISSN	Publisher	Abstract source
Journal of Finance	JF	0022-1082	Wiley	Web of Science
Review of Financial Studies	RFS	0893-9454	Oxford	Web of Science
Journal of Financial Economics	JFE	0304-405X	Elsevier	ScienceDirect API

Coverage: 2000–2024. Abstract text is available for 91.8% of *Journal of Finance* research articles, 97.7% of *Review of Financial Studies* articles, and 94.6% of *Journal of Financial Economics* articles after Elsevier API enrichment. The *Journal of Finance* denominator is restricted to research articles only; the raw Web of Science record for the *Journal of Finance* includes a large number of editorial matter, meeting announcements, and AFA administrative items (which carry no abstracts), so the overall entry-level coverage rate is 74.6%. Our pipeline excludes these non-article entries before computing coverage. Because the pipeline classifies NBER working papers using NBER abstracts—not journal abstracts—any residual *Journal of Finance* coverage gap does not affect our classification results.

3.2 NBER Working Papers

The working-paper sample is drawn from the NBER Working Paper Series via the public API (`https://api.nber.org`). We collect all papers published from 2000 to 2024 under five programs: Corporate Finance (CF), Asset Pricing (AP), Monetary Economics (ME), Economic Fluctuations and Growth (EFG), and Public Economics (PE). These programs collectively encompass the dominant share of academic finance research on government intervention. Papers appearing in multiple programs are counted once.

We use the NBER working paper sample as a reference distribution for comparison with the published sample. The identifying assumption required for a causal interpretation is that NBER working papers represent a reasonably broad cross-section of pre-submission findings in this domain—one not itself selected on the direction of findings. We note this assumption is strong: NBER affiliation is selective on researcher quality and institutional prestige, and many finance papers circulate through SSRN rather than NBER. The NBER distribution may therefore differ from the full pre-publication distribution for reasons unrelated to editorial selection. We return to this identification challenge formally in Section 4.1.

3.3 In-Scope Classification Pipeline

Identifying government-intervention papers in a corpus of 33,224 documents requires an automated approach. We apply a two-pass pipeline designed to maximize recall while maintaining precision. Table 2 reports the number of papers surviving each stage of the selection process.

Starting universe. We begin with all articles and working papers collected from our four sources—the *Journal of Finance*, the *Review of Financial Studies*, the *Journal of Financial Economics*, and the NBER Working Paper Series—for the period 2000–2024, yielding a combined corpus of 33,224 documents (25,801 NBER working papers plus 7,423 journal articles: 2,311 from the *Journal of Finance*, 2,426 from the *Review of Financial Studies*, and 2,686 from the *Journal of Financial Economics*).

Pass 1 — Keyword filter. A paper passes the first filter if its title or abstract contains at least one term from a pre-specified taxonomy of government intervention keywords spanning six categories: financial regulation (e.g., "Dodd-Frank," "capital requirements," "Basel"), securities law (e.g., "SEC," "Sarbanes-Oxley," "insider trading law"), banking policy (e.g., "FDIC," "deposit insurance," "TARP"), monetary and fiscal policy (e.g., "quantitative easing," "fiscal stimulus"), market microstructure rules (e.g., "circuit breaker," "tick size"), and corporate governance law (e.g., "say-on-pay," "proxy access," "mandatory disclosure"). Ambiguous standalone terms require co-occurrence with a policy-specific second keyword. The full taxonomy is reported in Appendix A. The keyword filter reduces the corpus from 33,224 documents to **900 candidates** (454 NBER, 110 *Journal of Finance*, 147 *Review of Financial Studies*, and 189 *Journal of Financial Economics*).

Pass 2 — LLM classifier. Each keyword-flagged candidate is passed to Claude (Anthropic, 2024) with the following instruction (full prompt in Appendix B):

> *"A finance research paper is 'in scope' if and only if its PRIMARY research question is the CAUSAL EFFECT of a specific government law, regulation, or public policy on financial markets, firms, or investors. Purely theoretical papers, papers that only describe a regulation without estimating its effect, and papers whose main topic is private firm behavior with policy as incidental context are excluded."*

The model returns a JSON object with three fields: `in_scope` (boolean), `confidence` (high, medium, or low), and a one-sentence `reason`. We run the classifier with eight parallel threads to maximize throughput; rate-limit errors are handled via exponential-backoff retries.

Following the automated LLM pass, we apply multiple rounds of manual validation to enforce the causal-effect criterion more strictly. The most consequential step removes papers that use a regulatory event *only* as an exogenous instrument to identify a non-regulatory relationship—for example, using SOX passage to instrument for disclosure quality in order to study its effect on firm investment. Such papers' primary research question is not the policy effect itself; they fall under the third LLM exclusion criterion (main topic is private behaviour with policy as incidental context) but were not consistently flagged by the automated classifier. The automated pipeline and manual validation together retain **200 in-scope papers** from the 900 candidates, an effective pass rate of 22.2%. The final analysis sample consists of **200 papers**: 68 NBER working papers and 132 published journal articles (27 in the *Journal of Finance*, 48 in the *Review of Financial Studies*, and 57 in the *Journal of Financial Economics*).

Table 2. Sample Selection

Selection step	Papers remaining	Papers dropped
Starting universe (2000–2024)		
NBER working papers	25,801	
Journal articles: JF	2,311	
Journal articles: RFS	2,426	
Journal articles: JFE	2,686	
Total corpus	**33,224**	
Pass 1: Keyword filter		
Retain keyword-hit papers	900	32,324
of which: NBER	454	
of which: JF	110	
of which: RFS	147	
of which: JFE	189	
Pass 2: LLM scope classification		
Retain in-scope papers	200	700
Final analysis sample		
Published in JF / RFS / JFE	132	
of which: JF	27	
of which: RFS	48	
of which: JFE	57	
NBER working papers (unpublished)	58	
Total	**200**	

Notes. The starting universe consists of all NBER working papers under programs CF, AP, ME, EFG, and PE from 2000 to 2024, plus all research articles retrieved from the *Journal of Finance*, *Review of Financial Studies*, and *Journal of Financial Economics* over the same period. Pass 1 applies a keyword taxonomy of government intervention terminology (Appendix A); Pass 2 uses a large language model (Claude) to verify that each flagged paper's primary research question is the causal effect of a specific government policy on financial markets, firms, or investors. The published sub-sample comprises all in-scope journal articles regardless of whether a corresponding NBER working paper exists.

3.4 Direction Coding

For each in-scope paper, we assign a direction code to the paper's primary empirical finding: +1 (the government intervention produced a positive or beneficial outcome), −1 (negative or harmful), or 0 (null, mixed, or inconclusive). Papers with multiple interventions are coded on the primary intervention identified in the title or opening sentence of the abstract.

The coding prompt instructs the model to code the *sign of the estimated coefficient* on the primary intervention, not the authors' normative framing. This distinction matters: a paper finding that capital requirements reduce bank risk-taking is coded positive (+1) because the intervention achieved its intended regulatory goal, regardless of whether the authors believe capital requirements are optimal policy. The full coding prompt is in Appendix B.

Coding reliability. We validate the direction codes through two exercises.

LLM replication (inter-rater reliability). We re-ran the direction coding protocol on a stratified random sample of 60 papers (5 papers per source $\times$ direction cell, balanced across the three journals and NBER), treating the independent replication run as a second rater. The two runs agree on 43 of 60 papers (71.7%). The linear-weighted Cohen's κ across all three direction categories is $\kappa = 0.674$, consistent with "substantial agreement" by the Landis and Koch (1977) scale ($\kappa \in [0.61, 0.80]$). Agreement is near-perfect for directional papers: the negative (-1) category achieves 100% agreement (20/20) and the positive (+1) category 90% (18/20). Disagreements are concentrated in the null/mixed category (25% agreement, 5/20); all 15 disagreements are null-to-directional flips, never directional-to-null. This asymmetry implies that any coding error attenuates our estimated premium, making coefficients a conservative lower bound. Appendix Table A3 reports the full confusion matrix.

Human validation. To assess agreement between LLM codes and a human standard, we drew a stratified random sample of 40 papers (20 published, 20 NBER) and asked the authors to independently code the direction of each paper's primary finding using only the abstract, without reference to the LLM output. The human coding sheet and comparison script (`scripts/13_human_validation_kappa.py`) are included in the replication package. Human-coded results and the corresponding κ statistics are reported in Table A4 in the online appendix.

3.5 Linking NBER Papers to Published Papers

Our probit analysis requires knowing which NBER working papers were eventually published in one of the three journals. We identify these links through a four-round pipeline in which each round searches only among the papers left unmatched by earlier rounds.

In the *first round* we apply token-sort Levenshtein similarity (RapidFuzz) to normalized titles, comparing each of the 132 journal papers against the full NBER archive of 30,219 working papers. A shared author surname is required as a second gate. Scores of 90 or above are automatically linked; scores in $[75, 90)$ are flagged for manual inspection by the authors. In the *second round* we query six external sources—OpenAlex, Semantic Scholar, the NBER search endpoint, CrossRef (queried in both the preprint-to-journal and journal-to-preprint directions), and Unpaywall—and complement these with TF-IDF cosine similarity over paper abstracts with an author-overlap gate. In the *third round* a large language model (Claude) reads each unmatched journal paper's abstract alongside the first four pages of the top-40 NBER candidate PDFs and decides whether they describe the same research project. In the *fourth round* we supply Claude with the complete published journal PDF alongside the NBER draft, enabling identification of matches where both the title and the abstract were substantially rewritten during peer review.

The pipeline identifies 10 matched pairs among the 132 in-scope journal articles. The majority of matches are identified in Round 1 via fuzzy title matching; Rounds 2–4 contribute the remainder. The remaining 122 journal articles (92.4%) are unmatched. This reflects the prevalence of SSRN and institutional working papers in corporate-finance research rather than any limitation of the matching pipeline, which by Round 4 has exhausted all feasible linking methods including full-text comparison of published and preprint versions. External verification via OpenAlex, Unpaywall, and CrossRef independently confirms that these 122 papers have no retrievable NBER preprint record.

To assess whether the matched and directly identified published papers are representative, we compare the direction codes of the 10 matched NBER papers with those of the 132 directly identified journal articles. Among matched NBER working papers, 30.0% have a directional finding; among the 132 directly identified journal articles, 75.0% do. This divergence likely reflects the shorter, less decisive NBER abstracts (44 words on average versus 104 for published abstracts) rather than a true direction difference: the LLM more readily assigns null codes to ambiguous NBER abstracts. The direction-code instability documented in Section 3.4 is consistent with this interpretation.

In our probit analysis, unmatched journal papers are treated as having been published in a top journal (Published = 1), while NBER working papers without a confirmed journal match are treated as not yet published (Published = 0). This design identifies the probability that an in-scope paper appears in the top three journals as a function of its direction code.

3.6 Summary Statistics

Table 3 presents summary statistics for the 200-paper analysis sample. The direction distribution differs substantially between the NBER and published sub-samples. Among NBER working papers, 75.0% are coded null or mixed, compared with 28.2% among published papers—a difference of 47 percentage points. Positive-finding papers constitute 31.7% and negative-finding papers 40.1% of the published sample, versus 10.3% and 14.7% of the NBER sample. These raw differences foreshadow the publication rate disparities documented in Section 5.

Table 3. Summary Statistics

Sample	N	Positive (%)	Negative (%)	Null (%)	Mean year
All papers	200	25.5	32.5	42.0	2014.7
NBER working papers	68	10.3	14.7	75.0	2012.6
Published (JF/RFS/JFE)	142	31.7	40.1	28.2	2015.6
By journal:					
Journal of Finance	27	33.3	29.6	37.0	2013.1
Review of Financial Studies	48	29.2	52.1	18.8	2017.1
Journal of Financial Economics	57	36.8	38.6	24.6	2015.9

Notes. The sample consists of 68 NBER working papers (programs CF, AP, ME, EFG, and PE, 2000–2024) and 132 published journal articles (*Journal of Finance*, *Review of Financial Studies*, *Journal of Financial Economics*) classified as in-scope government-intervention papers by our LLM pipeline and manual validation. Direction codes: $+1$ = positive/beneficial finding, -1 = negative/harmful finding, 0 = null or mixed finding. The NBER and Published rows are not mutually exclusive: 10 papers in the NBER working paper sample were matched to published journal articles and therefore appear in both the NBER ($N = 68$) and Published ($N = 142$) rows; the Published count of 142 is what enters the probit regressions as Published = 1. Column percentages may not sum to 100 due to rounding.

4. Empirical Strategy

4.1 Research Design and Identification

Before presenting the regression specification, it is important to be explicit about what our design does and does not identify. Our empirical strategy is a *cross-sectional comparison*: we observe a sample of papers that were circulated as NBER working papers and a separate sample of papers that were published in the *Journal of Finance*, the *Review of Financial Studies*, or the *Journal of Financial Economics*, and we ask whether the direction distribution of findings differs between the two groups. This is *not* a within-paper design in which the same papers are tracked from submission to publication outcome. Only 10 of 132 published papers (7.6%) are matched to an NBER working paper, so the two samples are largely non-overlapping populations.

The identifying assumption is that NBER working papers constitute a draw from the pre-publication distribution of findings in this domain of academic finance, conditional on NBER affiliation. Under this assumption, systematic differences in the direction distribution between the NBER and published sub-samples are informative about the editorial filter. This assumption is analogous to the design of Franco et al. (2014), who use NSF pre-registration to observe the pre-publication distribution of social-science findings; and to DellaVigna and Linos (2022), who obtain institutional RCT records. Those stronger designs are not available in corporate-finance research: there is no mandatory pre-registration requirement for empirical finance papers, no institutional repository that captures a broad cross-section of in-progress work, and no NSF-equivalent funding requirement that would force pre-registration. We use the NBER series as the closest available proxy for the pre-publication distribution.

Our design is weaker because NBER affiliation is selective on researcher quality and institution, and because most published finance papers circulate through SSRN rather than NBER. The direction of the resulting bias is ambiguous: NBER selection could make estimates *conservative* (if high-quality NBER researchers more often produce genuine null results, our null-paper baseline is unusually good, and the true gap for average researchers is larger); or it could make estimates *overstated* (if productive NBER researchers circulate work only once it is results-ready, the NBER corpus may be upwardly selected toward directional findings relative to all ongoing research, inflating the measured gap). We cannot resolve this ambiguity without submission-level data. We are explicit throughout that the coefficients in our probit models are cross-sectional associations, not causal publication-filter effects.

Limitation: the two samples are not the same papers at two points in the pipeline. Because only 7.6% of published papers are matched to an NBER counterpart, the NBER and journal samples are largely non-overlapping cross-sections selected by different institutional processes. The constructed "publication rate" of 71% is a function of the relative sizes of the two corpora the authors chose to assemble; it depends on NBER program coverage, abstract availability, and the decision to count unmatched NBER papers as published = 0. We do not claim to observe papers transitioning through the editorial filter; rather, we compare the direction distribution of pre-publication research (NBER working papers) to that of published articles, and ask whether the two distributions are consistent with a significance-sorting mechanism. This framing is analogous to Franco et al. (2014) and DellaVigna and Linos (2022), who likewise rely on cross-sectional comparisons between pre-registered and published samples. The key identifying assumption is that the NBER distribution is representative of the pre-publication distribution conditional on NBER affiliation, which we acknowledge is selective.

4.2 Primary Specification

Our primary test estimates a probit model at the paper level:

$$\Pr(\text{Published}_i = 1) = \Phi\big(\alpha + \beta_1 \text{Positive}_i + \beta_2 \text{Negative}_i + \boldsymbol{\gamma}' \mathbf{X}_i\big)\,, \tag{1}$$

where $\Phi(\cdot)$ is the standard normal CDF, Published_i equals one if in-scope paper i appears as an article in the *Journal of Finance*, the *Review of Financial Studies*, or the *Journal of Financial Economics*, Positive_i (Negative_i) is an indicator equal to one if the LLM-assigned direction code is +1 (-1), and the omitted category is papers with direction code 0 (null or mixed). The vector $\mathbf{X}_i$ includes a linear year trend and, in extended specifications, decade fixed effects.

Our primary coefficients of interest are β_1 and β_2. A finding of $\beta_1 > 0$ ($\beta_2 > 0$) indicates that positive-finding (negative-finding) papers are more likely to be in the published sample than null-finding papers, conditional on controls. Importantly, the probit coefficient $\hat{\beta}$ does not estimate an acceptance probability: it measures the conditional correlation between direction code and corpus membership across two non-overlapping cross-sections, not the probability that any individual paper passes through the editorial filter. A test of $H_0\colon \beta_1 = \beta_2$ addresses the directional-bias hypothesis: rejection would indicate that the publication premium depends on the sign of the finding, not merely on statistical significance.

4.3 Marginal Effects and Economic Interpretation

Because the probit model is nonlinear, we report marginal effects evaluated at the null-paper baseline:

$$\Delta_d = \Phi(\hat{\alpha} + \hat{\beta}_d) - \Phi(\hat{\alpha}), \quad d \in \{+, -\}.$$

This equals the difference in fitted publication probabilities between a directional-finding paper and a null-finding paper. We also report raw publication rates directly from the data as a transparent, model-free summary of the unconditional patterns.

4.4 Cross-Journal and Time-Trend Analysis

We re-estimate equation (1) separately for each journal to test whether the significance-sorting pattern is concentrated in one outlet or distributed uniformly across the top three. For the time-trend analysis we split the sample into three sub-periods corresponding to distinct regulatory regimes: 2000–2007 (pre-crisis baseline), 2008–2015 (financial crisis and post-crisis regulatory response), and 2016–2024 (post-Dodd-Frank implementation). We report results for all three sub-periods with appropriate caveats about sample size for the earliest period ($N = 28$).

5. Results

5.1 Directional vs. Null Findings

Table 4 reports the unconditional publication rates by direction code. Positive-finding papers are published at a rate of 88.2% (45 of 51 papers), and negative-finding papers at 87.7% (57 of 65). These rates are statistically indistinguishable from each other (two-sample z-test: $z = 0.09$, $p = 0.93$). By contrast, null-finding papers are published at only 47.6% (40 of 84), sharply below either directional category.

A χ^2 test of homogeneity across the three publication rates strongly rejects equality ($\chi^2 = 38.456$, $p < 0.001$). The null-finding publication rate of 47.6% implies that approximately

52% of in-scope papers finding no significant effect of government intervention do not appear in a top-three finance journal. This suppression of null results, if representative of the broader population of research, would substantially overstate the share of government interventions with detectable effects in the published record.

Table 4. Publication Rates by Finding Direction

	Sample composition				
Finding Direction	NBER-only	Published	Total	Publication Rate	95% CI
Positive (+1)	6	45	51	88.2%	[76.6, 94.5]
Negative (−1)	8	57	65	87.7%	[77.6, 93.6]
Null/Mixed (0)	44	40	84	47.6%	[37.3, 58.2]
Total	58	142	200	71.0%	

Notes. Direction-specific composition of the two-sample cross-section, 2000–2024. *Finding Direction* is the LLM-assigned sign of the paper's primary empirical result: +1 (positive/beneficial), −1 (negative/harmful), or 0 (null or mixed). *NBER-only* counts in-scope NBER working papers with no confirmed match to a JF/RFS/JFE article (Published = 0); *Published* counts in-scope journal articles (Published = 1); *Total* is the sum across both sub-samples. *Publication Rate* is $N_{\text{Published}}/N_{\text{Total}}$ within each direction category; it is a cross-sectional ratio, not an estimated acceptance probability. *95% CI* is the Wilson score confidence interval. A χ^2 test of homogeneity across the three rates gives $\chi^2 = 38.456$ ($p < 0.001$). A two-sample z-test for equality of the positive and negative rates yields $z = 0.09$ ($p = 0.93$).

5.2 Probit Regression Results

Table 5 reports estimates across six specifications. Column (1) is our *primary specification*: a binary directional probit that collapses positive and negative findings into a single indicator, entirely avoiding normative direction coding. Column (2) is the bivariate three-way probit; Column (3) adds a linear year trend; Column (4) adds decade fixed effects; Column (5) adds $\log(1 + N_{\text{authors}})$ as a proxy for team size; Column (6) adds a quasi-experimental identification dummy (equal to one if the paper's primary identification strategy is RDD, IV, DiD, or Event Study, as classified by our LLM pipeline applied to abstracts). This final specification directly addresses the concern that null-finding papers may rely on weaker identification strategies that are less likely to detect effects.

Column (1) shows that directional papers are 1.231^{***} probit units (s.e. 0.204) more likely to appear in a top journal than null papers—a 40 percentage-point gap. This is our cleanest result because it makes no normative judgment about the sign of the finding.

In Column (2), both $\hat{\beta}_1$ (Positive) and $\hat{\beta}_2$ (Negative) are large, positive, and highly significant: $\hat{\beta}_1 = 1.247$ (s.e. 0.267, $p < 0.001$) and $\hat{\beta}_2 = 1.219$ (s.e. 0.242, $p < 0.001$). A Wald test fails to reject equality of the two coefficients ($\chi^2 = 0.008$, $p = 0.93$), consistent with symmetry across directions. This test has limited power: the minimum detectable difference at 80% power is 0.851 coefficient units, so the non-rejection does not rule out moderate directional asymmetries. We present this as an absence of detected directional asymmetry, not as evidence that the publishing premium is symmetric. Adding the author-count quality proxy in Column (5) leaves the direction coefficients virtually unchanged ($\hat{\beta}_1 = 1.213^{***}$, $\hat{\beta}_2 = 1.137^{***}$), indicating that team size does not explain the null-paper disadvantage. The author-count coefficient is positive and marginally significant ($\hat{\delta} = 0.770^{*}$, s.e. 0.407). The estimated year trend (Column (3)) is positive and marginally significant ($\hat{\gamma} = 0.032^{*}$, s.e. 0.017), suggesting a modest upward drift in the publication–direction association over the period. In Column (5), the year trend becomes statistically insignificant ($\hat{\gamma} = 0.027$, s.e. 0.017) after controlling for team size. Decade fixed effects (Column (4)) leave the direction coefficients essentially unchanged.

Table 5. Direction–Publication Association: Probit Estimates

	(1) Binary (Primary)	(2) Bivariate	(3) + Year	(4) + Decade FE	(5) + Authors	(6) + ID Strategy
Directional	1.231***					
	(0.204)					
Positive (+1)		1.247***	1.198***	1.210***	1.213***	1.228***
		(0.267)	(0.269)	(0.268)	(0.270)	(0.281)
Negative (−1)		1.219***	1.111***	1.180***	1.137***	1.228***
		(0.242)	(0.251)	(0.250)	(0.255)	(0.270)
Year trend			0.032*		0.027	−0.001
			(0.017)		(0.017)	(0.013)
$\log(N_{\text{authors}})$					0.770*	0.013
					(0.407)	(0.313)
Quasi-Exp. ID						0.734***
						(0.221)
Decade FEs	No	No	No	Yes	No	No
Intercept	−0.060	−0.060	0.040	−0.060	−0.931*	−1.274**
N	200	200	200	200	200	200
Pseudo R^2	0.163	0.163	0.178	0.173	0.193	0.241

Notes. Probit estimates of the probability that an in-scope paper is published in the *Journal of Finance*, *Review of Financial Studies*, or *Journal of Financial Economics*. Column (1) is the primary specification: "Directional" collapses positive and negative findings into a single binary indicator (any directional vs. null), avoiding normative coding of the direction sign. Columns (2)–(6) use the three-way direction code; Positive (+1) and Negative (−1) are indicators for the LLM-assigned direction code; the omitted category is Null/Mixed (0). Column (5) adds $\log(1 + N_{\text{authors}})$ as a proxy for team size. Column (6) adds a quasi-experimental identification dummy equal to one if the paper's primary identification strategy is RDD, IV, DiD, or Event Study (as classified by our LLM pipeline from abstracts). Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Marginal effects: for Col. (1), at the binary-spec baseline ($\hat{\alpha} = -0.060$, implied null publication probability $\Phi(-0.060) \approx 47.6\%$), the directional gap is 40 pp. For Cols. (2)–(6), at the bivariate-spec baseline ($\hat{\alpha} = -0.060$, implied null publication probability $\approx 47.6\%$), 41 pp for positive-finding and 40 pp for negative-finding papers in Col. (2). A Wald test for $H_0\colon \hat{\beta}_1 = \hat{\beta}_2$ in Column (2) yields $\chi^2 = 0.008$, $p = 0.93$. At the observed sample size the test has only 5.9% power; the minimum detectable difference at 80% power is 0.864 coefficient units. Failure to reject does not rule out moderate directional asymmetries.

5.3 Economic Magnitude

The probit marginal effects imply substantial economic magnitudes. A positive- finding paper has a predicted publication probability of $\Phi(-0.060 + 1.247) = \Phi(1.187) = 88.2\%$, compared with $\Phi(-0.060) = 47.6\%$ for a null- finding paper (bivariate-spec baseline)—a difference of 41 percentage points. Using raw publication rates, directional-finding papers are approximately 1.85 times as likely to appear in a top-three journal as null-finding papers: the ratio for positive findings is $88.2\%/47.6\% = 1.85$; for negative findings, $87.7\%/47.6\% = 1.84$.

These magnitudes exceed analogous estimates from other literatures. Brodeur et al. (2016) estimate that the relative likelihood of a significant versus borderline-insignificant result appearing in print in top economics journals is roughly 1.5–2.0. Our $\approx 1.85\times$ ratios suggest that the finance-regulation literature is associated with a particularly pronounced significance-sorting pattern, plausibly reflecting the direct policy relevance of this research domain and corresponding high readership value placed on papers with actionable, directional conclusions. Whether this association reflects editorial selection, author self-selection in submission, or both cannot be determined from our design.

5.4 Cross-Journal Analysis

Table 6 reports journal-specific probit estimates. The significance-sorting pattern is strong and statistically significant in the *Review of Financial Studies* and *Journal of Financial Economics*. The *Journal of Financial Economics* exhibits the largest positive coefficient ($\hat{\beta}_1 = 1.448^{***}$, $p < 0.001$), and the *Review of Financial Studies* the largest negative coefficient ($\hat{\beta}_2 = 1.459^{***}$, $p < 0.001$). Taken together, these two journals drive the pooled result.

The *Journal of Finance* results are positive, statistically significant, and in the same direction as the other journals ($\hat{\beta}_1 = 1.150^{***}$, $p < 0.001$; $\hat{\beta}_2 = 0.900^{**}$, $p < 0.05$), but *smaller in magnitude* than those of the *Review of Financial Studies* and *Journal of Financial Economics*. Cross-journal Wald tests for equality of the positive coefficients yield $p = 0.34$ (JF vs. JFE), $p = 0.63$ (JF vs. RFS), and $p = 0.64$ (RFS vs. JFE); none of the pairwise differences is statistically distinguishable. We caution strongly that these tests have low power: the JF sub-sample contains only 27 in-scope published articles, making cross-journal comparisons highly underpowered. The point estimates are consistent with a smaller direction–publication association at the *Journal of Finance*, but this difference cannot be statistically confirmed at the available sample sizes.

Table 6. Probit Models by Journal

	Journal of Finance	*Review of Financial Studies*	*Journal of Financial Economics*
Positive (+1)	1.150***	1.287***	1.448***
	(0.383)	(0.377)	(0.329)
Negative (−1)	0.900**	1.459***	1.232***
	(0.382)	(0.333)	(0.312)
Year trend	−0.001	0.044*	0.051**
	(0.024)	(0.025)	(0.022)
N	85	106	115
Pseudo R^2	0.114	0.273	0.236

Notes. Each column re-estimates equation (1) treating publication in the specified journal as the outcome (versus any of the remaining two journals or the NBER-only group). Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The *Journal of Finance* sample contains only 27 in-scope published articles in 2000–2024, limiting statistical power for journal-specific tests.

5.5 Time-Period Heterogeneity (Exploratory)

The following sub-period analysis is exploratory. Sub-sample sizes are small ($N = 28$, 69, and 103 for the three periods) and the results should be interpreted as descriptive evidence, not confirmatory tests. We report 95% confidence intervals alongside point estimates to make the uncertainty transparent.

Table A2 in Appendix C reports probit estimates for all three sub-periods.

During the *pre-crisis period* (2000–2007, $N = 28$), the positive- finding premium ($\hat{\beta}_1 = 1.600^{**}$, 95% CI [0.370, 2.831]) is large and significant, while the negative-finding premium ($\hat{\beta}_2 = 0.319$, 95% CI [−1.528, 2.165]) is imprecisely estimated. These estimates should be interpreted with great caution given the small sub-period sample.

During the *financial crisis and post-crisis period* (2008–2015, $N = 69$), the negative-finding premium ($\hat{\beta}_2 = 1.307^{***}$, 95% CI [0.528, 2.086]) was somewhat larger than the positive-finding premium ($\hat{\beta}_1 = 1.148^{**}$, 95% CI [0.193, 2.103]), though the confidence intervals overlap substantially. Whether this pattern reflects editorial preferences, author self-selection in submission, or the broader research environment during the post-crisis regulatory period cannot be determined from these data.

In the *post-Dodd-Frank period* (2016–2024, $N = 103$), the two premiums converge markedly: positive findings carry a premium of 0.995^{**} (95% CI [0.237, 1.753]) and negative findings 1.022^{***} (95% CI [0.333, 1.712]). The overlapping intervals are consistent with a more symmetric direction-publication association in the post-implementation period, though the small sub-period samples preclude strong inference about the source of the earlier divergence.

6. Robustness

Table 7 summarizes the robustness checks. We organize them into four groups: sample and scope, alternative specifications, classification and quality concerns, and an alternative pre-publication baseline.

6.1 Sample and Scope Robustness

Matched pairs only. Among the 25 published papers with a confirmed NBER working-paper match, we independently LLM-code the direction from the NBER *draft* abstract and from the published abstract, yielding 50 observations (25 draft-stage, 25 published-stage) for the same set of papers. Direction codes were stable across versions in 15 of 25 pairs (60%); the remaining pairs shifted predominantly from null at draft stage to directional at publication stage, consistent with attenuation bias in the main NBER control group. The binary probit on this matched sample yields $\hat{\beta} = 0.935^{**}$ ($N = 50$, $p = 0.011$), qualitatively consistent with the main estimate.

Excluding monetary policy papers. We exclude the 20 papers whose primary topic is monetary or central bank policy (quantitative easing, discount window, lender-of-last-resort operations, and unconventional monetary policy) and restrict attention to explicit financial legislation, securities regulation, and banking supervision. The results remain highly significant: $\hat{\beta}_1 = 0.999^{***}$ and $\hat{\beta}_2 = 1.018^{***}$ ($p < 0.001$, $N = 180$), confirming that the significance-sorting pattern is not driven by the inclusion of monetary policy papers.

Journal subsets. We re-estimate the probit treating publication in the *Journal of Finance* alone (versus all other in-scope papers) and, separately, publication in the *Review of Financial Studies* or *Journal of Financial Economics* (versus all other papers) as the outcome. Both premiums are statistically insignificant in the *Journal of Finance*-only specification ($\hat{\beta}_1 = 0.297$, $\hat{\beta}_2 = 0.135$, $p > 0.10$), consistent with the low-power caveat noted in Section 5. The *Review of Financial Studies*+*Journal of Financial Economics* specification yields large and highly significant coefficients ($\hat{\beta}_1 = 1.037^{***}$, $\hat{\beta}_2 = 1.038^{***}$, $p < 0.001$), confirming that the overall result is driven by these two journals.

6.2 Alternative Specifications

Binary directional indicator. A potential concern about our direction coding is that classifying a finding as positive versus negative rests on a normative judgment about what constitutes a beneficial intervention outcome. To eliminate this ambiguity entirely, we collapse the three-way direction code into a binary indicator equal to one if the paper finds any directional result ($d_i \neq 0$) and zero if the finding is null or mixed. The coefficient is 1.231^{***} (s.e. 0.204, $p < 0.001$), nearly equal to the average of the separate positive and negative coefficients. Adding the year trend and author-count control leaves the coefficient essentially unchanged at 1.173^{***} (s.e. 0.212), confirming that the result does not hinge on any particular normative coding convention.

Linear probability model. Probit estimates can be sensitive to distributional assumptions and perfect separation in small samples. Re-estimating the main specification as a linear probability model (OLS with heteroskedasticity-robust standard errors) yields coefficients of 0.406*** (s.e. 0.072) for positive findings and 0.401*** (s.e. 0.069) for negative findings, directly interpretable as percentage-point differences in publication probability relative to null papers. These magnitudes closely match the probit marginal effects and confirm the results do not hinge on the probit functional form.

NBER-to-journal matching sensitivity. We verify that results hold when we drop all matched NBER-journal pairs and estimate the model on papers with unambiguously clear publication status. The resulting coefficients are nearly identical to the full-sample estimates, confirming that the results do not hinge on the small matched subsample.

6.3 Classification and Quality Concerns

Quasi-experimental papers only. The most substantive quality-control check restricts the sample to papers using quasi-experimental identification strategies (RDD, IV, DiD, or event study), as classified by our LLM pipeline from abstracts. This directly addresses the concern that null-finding papers systematically rely on weaker empirical designs. Among quasi-experimental papers ($N = 100$), the positive coefficient remains large and significant ($\hat{\beta}_1 = 1.603^{***}$); the negative coefficient cannot be identified due to perfect separation (all quasi-experimental negative-finding papers in the sample are published). This confirms that the significance-sorting result is not driven by an identification-quality gradient between directional and null papers.

Abstract language and LLM classification confidence. Abstract polish introduces a two-sided measurement concern. Among published papers, 75.4% receive a high-confidence direction code; among NBER-only papers, only 15.5% do, with 74.1% receiving low-confidence ratings. On one hand, hedged NBER abstract language may push genuinely directional papers into the null category, attenuating the estimated premium. On the other hand, if authors revise abstracts to sound more decisive after a revise-and-resubmit, the LLM may code published papers as directional partly because their language signals directionality more clearly, inflating the premium.

The cleanest test of whether abstract-length asymmetry drives the finding is a symmetric truncation: we recode all direction codes from only the first 100 words of each abstract, equalising the text available to the LLM across both corpora. Published abstracts average 94 words after truncation versus 45 for NBER papers (the truncation is binding only for published papers), so the window is still not perfectly symmetric, but it eliminates the bulk of the length gap. The results are qualitatively unchanged: $\hat{\beta}_1 = 1.179^{***}$ ($p < 0.001$) and $\hat{\beta}_2 = 1.350^{***}$ ($p < 0.001$), reported in Table 7 Panel C. Abstract length therefore cannot account for the direction-publication association.

Our replication exercise (Section 3.4) further supports this conclusion: of the 20 null-coded papers in the validation sample, 15 are reclassified as directional in the independent replication run, while no directional-coded papers are reclassified as null. This asymmetry implies that null-coding error attenuates rather than inflates our estimate.

Table 7 Panel C also reports a bounding specification that adds log(abstract length) as a covariate ($\hat{\beta}_1 = 0.171$, $p = 0.766$; $\hat{\beta}_2 = 0.322$, $p = 0.486$). This bounds the association from below: because abstract length is partly shaped by journals requiring polished abstracts at acceptance, the control is endogenous (close to a second proxy for publication status), so this coefficient is a lower bound rather than an unbiased estimate. Together, the main uninstrumented estimates provide an *upper bound* on the direction-publication association if abstract polish fully drives it, and the abstract-length control provides an *extreme lower bound* if all abstract-length variation is editorial. The symmetric 100-word truncation sits between these poles and does not share the endogeneity problem: its results ($\hat{\beta}_1 = 1.179^{***}$, $\hat{\beta}_2 = 1.350^{***}$) indicate that the true association is well above the lower bound.

Matched-pairs direction stability. Among the 25 papers confirmed to exist as both an NBER working paper and a top-journal publication, we independently LLM-code direction from the NBER draft abstract and from the published abstract. Draft-stage codes are directional for 32.0% of pairs versus 68.0% at publication stage; direction codes are stable across versions in 15 of 25 pairs (60%). The 10 pairs that shift do so predominantly from null at draft stage to directional at publication—consistent with the attenuation pattern documented above. Despite the small sample ($N = 50$ observations), the binary probit on the matched pairs yields $\hat{\beta} = 0.935^{**}$ ($p = 0.011$), qualitatively consistent with the main estimate. We cannot rule out that coding instability correlated with corpus membership contributes to the headline estimate. The symmetric truncation and matched-pairs checks together bound the concern from two directions: the truncation removes the abstract-length channel and the matched-pairs analysis holds the paper fixed.

6.4 AFA Conference Baseline

The central identification concern is that NBER working papers and top-journal articles are drawn from fundamentally different populations: only 7.6% of published papers are matched to an NBER counterpart, so the two samples are largely non-overlapping cross-sections that may differ in quality, topic mix, and institutional origin. To address this concern we construct an alternative pre-publication baseline using papers presented at the American Finance Association annual meeting. The test is designed to identify the *stage* at which the significance premium arises rather than to provide an additional quality screen. AFA papers represent completed research with final empirical results that have been publicly presented—they are pre-journal submissions in every meaningful sense. If the premium reflects pre-existing quality differences between directional and null papers (a confound independent of editorial selection), it should be visible at the AFA stage too. If it arises from journal editorial review, it should appear only after papers enter that process.

We also note that the quality-confound concern is limited by the nature of NBER affiliation itself. Posting to the NBER Working Paper series requires affiliation with the NBER research network, which is restricted to faculty at leading research universities—an elite and selective credential in academic economics. Null-finding NBER papers do not come from a marginal or unknown researcher pool; the 40-percentage-point publication gap within this uniformly elite group is difficult to attribute to quality differences alone.

We collect 4,080 AFA program papers from 2006–2024. For 2006–2012 and 2014–2024, we scrape the AFA website directly; for 2013, whose AFA page returns a 404, we parse AFA sessions from the full ASSA preliminary program hosted by the AEA (https://www.aeaweb.org/conference/2013/preliminary.php), yielding 186 papers from 57 AFA sessions. We supplement AFA conference titles with abstracts retrieved from OpenAlex, CrossRef, and the NBER working paper dataset (local fuzzy-title matching), obtaining abstracts for 3,960 of the 3,972 AFA papers (99.7% overall; coverage is higher for in-scope papers because government-intervention papers are more likely to be published). We apply our standard title-plus-abstract keyword filter (identical to the NBER pipeline) to papers with abstracts, and the broader title-only filter to the remaining papers, yielding 108 in-scope papers. Direction codes are assigned as follows: papers matched to our published dataset receive the direction code from the LLM-based abstract coding in our main pipeline; papers with available abstracts but no published match receive abstract-based keyword-rule codes; and the remaining papers (those without abstracts and unmatched) receive conservative title-keyword rules, which predominantly yield null codes because finance titles are typically neutral. 29 AFA papers (26.9%) are matched to a JF/RFS/JFE publication via fuzzy title matching.

Coding limitation. A known limitation of the AFA test is that direction codes are *not* assigned by identical methods across samples: the NBER and published samples use LLM abstract-coding throughout, while unmatched AFA papers with abstracts use keyword rules and papers without abstracts use title rules. Because keyword and title rules default more often to null when abstracts are ambiguous or absent, this asymmetry likely *overstates* the null share of the AFA sample and therefore *understates* the AFA directional premium relative to a uniform-LLM-coded baseline. The absence of a premium at the AFA stage ($\hat{\beta} = 0.077$) can therefore be interpreted as an upper bound on the true AFA premium: if anything, uniform LLM coding would raise this estimate. An analysis using LLM coding uniformly across all three samples is presented in the online appendix.

The binary directional probit on the AFA baseline yields $\hat{\beta} = 0.077$ (s.e. 0.298, $N = 108$), statistically indistinguishable from zero and far below the NBER-baseline estimate of 1.231^{***} (s.e. 0.204). Directional AFA papers are published in top journals at a 27.5% rate (22 of 80) versus 25.0% for null-finding papers (7 of 28). The pattern is unchanged when a year trend is added ($\hat{\beta} = 0.082$, s.e. 0.300). Direction coding applies the same three-tier hierarchy as the NBER pipeline: journal-matched papers receive direction codes from published-abstract coding; papers with OpenAlex, CrossRef, or NBER abstracts receive abstract-based keyword rules; and the remaining papers receive conservative title-keyword rules. The absence of a directional premium at the AFA stage, against the large premium in the NBER sample, localizes the significance-sorting mechanism to the journal editorial review process. This is precisely what a publication-bias interpretation predicts: completed research at the pre-journal stage shows no direction-based sorting; the premium emerges only at the publication stage. A pure quality-confound story predicts the opposite—quality differences between directional and null papers should be visible before journal review, not arise within it.[2]

Quasi-experimental papers only. The most substantive quality-control check restricts the sample to papers using quasi-experimental identification strategies (RDD, IV, DiD, or Event Study), as classified by our LLM pipeline applied to abstracts. This directly addresses the concern that null-finding papers systematically rely on weaker identification strategies. Among quasi-experimental papers ($N = 100$), the positive coefficient remains large and statistically significant ($\hat{\beta}_1 = 1.603^{***}$); the negative coefficient cannot be identified due to perfect separation (all quasi-experimental negative-finding papers are published). This confirms that the result is not driven by an identification-quality gradient between directional

[2]As a direct test of overlap, we identified 22 papers appearing in both the NBER and AFA datasets via fuzzy title matching (token-sort score ≥ 90). In this subsample, directional papers publish at 81.2% versus 66.7% for null papers (LPM $\hat{\beta} = 0.146$, s.e. 0.208, $p = 0.49$, $N = 22$). The test is severely underpowered (only six null papers in the overlap), so no strong inference is possible. The 14 pp gap being smaller than the 40 pp NBER gap is consistent with the AFA screen selecting papers already close to the published pool—it is not evidence that quality confounding drives the NBER result.

and null papers.

Linear probability model. Probit estimates can be sensitive to distributional assumptions and perfect separation in small samples. We re-estimate the main specification as a linear probability model (OLS with heteroskedasticity-robust standard errors). The LPM yields coefficients of 0.406^{***} (s.e. 0.072) for positive findings and 0.401^{***} (s.e. 0.069) for negative findings, directly interpretable as percentage-point differences in publication probability relative to null-finding papers. These magnitudes closely match the probit marginal effects (41 pp and 40 pp) and confirm the results do not hinge on the probit functional form.

Table 7. Robustness Checks

Specification	Variable	Coefficient	p-value	N
Panel A: Sample and Scope				
Excl. monetary policy	Positive	0.999***	<0.001	180
	Negative	1.018***	<0.001	180
Journal of Finance only	Positive	0.297	0.283	200
	Negative	0.135	0.629	200
Review of Financial Studies+Journal of Financial Economics only	Positive	1.037***	<0.001	200
	Negative	1.038***	<0.001	200
Matched pairs only	Directional	0.935**	0.011	50
[N=25 paper pairs]	Draft vs. published version of same paper			
Panel B: Alternative Specifications				
Binary directional	Directional	1.231***	<0.001	200
Binary + year + authors	Directional	1.173***	<0.001	200
LPM (OLS, HC3)	Positive	0.406***	<0.001	200
	Negative	0.401***	<0.001	200
Panel C: Classification and Quality Concerns				
Quasi-exp. papers only	Positive	1.603***	<0.001	100
	Negative	separation		100
100-word truncation	Positive	1.179***	<0.001	200
[symmetric window]	Negative	1.350***	<0.001	200
+ log(abstract length)	Positive	0.171	0.766	200
[lower-bound spec]	Negative	0.322	0.486	200
Panel D: Alternative Pre-Publication Baseline				
AFA conference baseline	Directional	0.077	0.797	108
(+ year trend)	Directional	0.082	0.785	108
Uniform LLM coding	Directional	0.038	0.900	108
[AFA, + year]	Directional	0.044	0.884	108

Notes. Each row reports the probit (or OLS for the LPM specification) coefficient on the indicated variable. The omitted category is null/mixed ($d_i = 0$), except for binary directional specifications where it is null/mixed vs. any directional finding. All probit specifications include a linear year trend unless otherwise noted. "Matched pairs only" uses the 25 papers confirmed to exist in both NBER and published form; the LLM codes each paper's direction from its NBER *draft* abstract and from its published abstract separately, yielding $N = 50$ observations (25 draft, 25 published); direction codes were stable across versions in 60% of pairs. "Binary + year + authors" adds $\log(1 + N_{\text{authors}})$ to the binary specification. The quasi-experimental restriction uses identification strategies classified by our LLM pipeline from abstracts ($N = 100$). The 100-word truncation specification re-codes all paper directions from the first 100 words of the abstract only, creating a symmetric text window across pre-publication and published papers. The abstract-length specification is a lower-bound bounding exercise: the control is partly endogenous (journals require longer abstracts during revision), so these coefficients represent a lower bound on the true direction-publication association. The AFA conference baseline uses AFA annual meeting papers (2006–2024, $N = 108$ in-scope) as the pre-publication comparison group; direction codes follow the same three-tier hierarchy as the NBER pipeline. "Uniform LLM coding" replaces keyword-based direction codes for unmatched AFA papers with the same LLM pipeline used for the main NBER/published sample; $\beta \approx 0$ is unchanged, confirming the AFA stage-identification result is not an artifact of the mixed coding method. Standard errors omitted for brevity. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

7. Discussion

7.1 Policy Implications

Our results are *consistent with* the published record of government-intervention research in finance overstating the proportion of interventions with statistically detectable effects. To illustrate the potential magnitude—this is illustrative arithmetic, not a quantified distortion—suppose the true distribution of findings were one-third positive, one-third negative, and one-third null. The published distribution would then show positive and negative findings at roughly $88\% \times 33\% = 29\%$ each, and null findings at only $48\% \times 33\% = 16\%$—less than half their true prevalence. Meta-analyses that take the published distribution at face value may overstate how often government intervention in financial markets has measurable effects. Whether this distortion is large or small depends on the true underlying distribution, which our design cannot identify.

The policy implications of the pattern depend entirely on which mechanism generates it, and our design cannot determine this. If the null-paper disadvantage is driven primarily by researcher self-censorship (*supply-side*), editorial interventions such as registered reports or results-blind review may have limited effect; addressing author incentives to write up and submit null results would be more important. If the mechanism is primarily *demand-side* (editors and referees differentially advancing directional papers), then editorial policy changes are more directly responsive. Our design cannot distinguish between these channels. What we can say is that, regardless of mechanism, the net result is the same: null results in this domain appear systematically underrepresented in the published record. Structural bias-correction methods (Andrews and Kasy, 2019) applied to specific intervention literatures (capital requirements, short-sale bans, mandatory disclosure) could provide regulators with plausibly unbiased assessments of policy efficacy, whether the underlying cause is supply- or demand-side.

7.2 Comparison to Other Domains

Our $\approx 1.85\times$ directional-to-null publication ratio is at the upper end of analogous estimates from adjacent literatures. Brodeur et al. (2016) document excess mass just above conventional significance thresholds in the *American Economic Review*, the *Journal of Political Economy*, and the *Quarterly Journal of Economics*, with implied ratios of roughly 1.5–2.0$\times$ for significant versus borderline-insignificant findings. Franco et al. (2014), studying NSF-funded social science experiments with pre-registered designs, find that 65% of null results were never published versus 21% of significant results—also implying a roughly 3$\times$ publication premium. DellaVigna and Linos (2022) find, in a sample of nudge-unit RCTs with institutional outcome tracking, that studies with detectable effects are substantially more likely to appear in print than null-result studies. Brodeur et al. (2023) document that pre-registration and pre-analysis plans reduce but do not eliminate these biases in economics.

Our $\approx 1.85\times$ ratio thus sits near the upper end of these estimates, which is consistent with the hypothesis that government-intervention research—a domain where policy implications are especially salient—exhibits a particularly pronounced significance-sorting pattern. Whether the bias in this domain exceeds that in comparable finance sub-fields without explicit policy intervention (e.g., corporate governance, asset pricing) is an important question for future research.

7.3 Mechanisms

7.4 What the Data Can and Cannot Tell Us

Our empirical design documents a cross-sectional association between finding direction and publication status. This association is consistent with two broad classes of mechanism, and distinguishing them is important because they have different policy implications. Our data cannot distinguish between them, and we do not attempt to do so.

Two broad classes of mechanism could generate the significance-sorting pattern we document, and it is important to distinguish them because they have different implications for policy.

Demand-side mechanisms. Editors and referees may treat statistical significance as a quality or novelty signal, disproportionately advancing papers with directional conclusions through the review process (*editorial selection*). Alternatively, specification search by authors—trying additional subsamples, controls, or outcome variables until a significant coefficient emerges—can produce a false impression of significance that survives peer review (Brodeur et al., 2016). Both of these are demand-side mechanisms in the sense that the editorial process plays an active filtering role.

Supply-side mechanisms. Researchers may be less likely to write up and submit null results to top journals, anticipating rejection and preferring to allocate effort to papers with directional findings (*strategic non-submission*). Franco et al. (2014) document that this behavior is common in the social sciences; of NSF-funded projects with null results, 65% were never written up for submission. Under a pure supply-side story, journals passively reflect whatever mixture of results is submitted to them, and the observed pattern results from rational author behavior rather than editorial bias.

What the data can and cannot tell us. Our design cannot distinguish between these mechanisms. Both generate identical observable predictions: null results underrepresented in the published record. The exploratory sub-period heterogeneity documented in Section 5.5—the surge in the negative-finding premium during 2008–2015 and its subsequent convergence—is consistent with a demand-side mechanism, because changing policy conditions plausibly shifted editorial appetites. However, the same pattern is equally consistent with researchers strategically responding to perceived editorial demand signals, or with differential submission behavior across periods. We do not treat this time variation as evidence for either mechanism.

Without submission-level data (desk-rejection rates by direction, time-to-first- decision, share of null-result papers sent out for review), we cannot directly test these channels. Future work following the approach of DellaVigna and Linos (2022)—who obtain institutional data on submission and publication outcomes for nudge-unit RCTs—would allow a sharper test.

7.5 Limitations

Five main threats to the validity of our analysis deserve discussion.

Fundamental identification constraint. The most important limitation is that NBER working papers and published journal articles are not different states of the same population of papers. They are largely non-overlapping samples selected through different institutional processes: only 10 of 132 published papers (7.6%) are matched to an NBER working paper. Our design therefore compares two different cross-sections, not papers tracked through the editorial filter. The observed difference in direction distributions is consistent with editorial selection but equally consistent with NBER papers having a different direction distribution for reasons unrelated to journals—for example, NBER researchers may strategically circulate papers with more ambiguous findings, or null-finding papers in this domain may disproportionately circulate through SSRN rather than NBER. Addressing this fully would require either submission-level data from journal editors (desk-rejection rates and review outcomes by direction code) or a pre-registration registry for finance papers analogous to the AEA RCT Registry. SSRN preprint histories for authors in our sample would also allow a more comprehensive pre-publication baseline by capturing papers that circulate outside NBER. We interpret all coefficients as cross-sectional associations, not causal editorial-filter effects.

LLM classification errors and abstract polish. Our pipeline assigns direction codes by reading paper abstracts. A key concern—confirmed by our data—is that published paper abstracts are substantially more detailed and decisive in their language than NBER working paper abstracts: the mean published abstract contains 104 words, compared with 45 words for NBER abstracts. Accordingly, 75.4% of published papers receive a "high-confidence" LLM direction code, versus only 15.5% of NBER-only papers. If the LLM more often assigns a null code to ambiguously phrased NBER abstracts, the measured null-paper publication rate would be biased downward—which would *attenuate* our estimated significance premium rather than inflate it.

Our replication exercise (Section 3.4) provides direct evidence on this attenuation: of the 20 null-coded papers in the validation sample, 15 are re-classified as directional in the independent replication run, confirming that the null category is the least stable. This instability *reduces* the significance premium we estimate, because some papers we code as null would be coded as directional by an alternative rater—which would increase both the numerator and denominator of the directional publication rate but reduce the denominator of the null rate. Our reported estimates should thus be interpreted as a conservative lower bound on the true bias. Future work with standardized abstract templates, full-paper coding (rather than abstract coding), or pre-registered abstract language could address this concern directly.

Quality controls and the identification-strategy alternative. A natural alternative explanation for the null-paper disadvantage is that null-finding papers rely on weaker empirical designs that are less likely to detect effects regardless of the true effect size. We address this through two controls. First, Spec (4) adds $\log(1 + N_{\text{authors}})$: the direction coefficients are virtually unchanged and the author-count coefficient is positive and marginally significant ($\hat{\delta} = 0.770^{*}$, s.e. 0.407), suggesting that larger teams are more likely to publish—an independent quality gradient that does not absorb the direction premium. Second, and more directly, Spec (5) adds a quasi-experimental identification dummy (equal to one if the paper uses RDD, IV, DiD, or Event Study methods, as classified by our LLM pipeline from abstracts). Papers with quasi-experimental designs are both more likely to detect genuine effects *and* more likely to be published; including this control absorbs the quality gradient that might be confounded with direction coding. The direction coefficients remain large and highly significant in Spec (5), confirming that the significance premium is not an artifact of identification quality.

We acknowledge that the LLM-based identification-strategy coding from abstracts is itself imperfect: fewer than half of abstracts explicitly name their identification strategy, so the LLM infers it from other signals. A hand-coded quality assessment across all 200 papers—using full paper text to classify identification strategy, sample size, and data quality—would provide a more definitive quality control and is a natural extension of this work.

NBER selection and the direction of potential bias. NBER researchers are among the most productive and well-connected academics in finance. If their null-finding papers are more publishable than null- finding papers from the broader population, our 47.6% null-paper publication rate overstates the true rate for the average researcher. This selection concern makes our significance-sorting estimate *conservative*: the true publication gap for directional versus null results across all finance researchers would be larger than we document. NBER selection—if anything—biases against finding a significance-sorting pattern, reinforcing rather than undermining our main conclusion. Absent submission-level data, the most reassuring feature of our sample is the low match rate (7.6%), which reflects broad SSRN circulation in this domain, and we cannot follow the same papers through the editorial filter as Franco et al. (2014) do for NSF-funded social science projects or as DellaVigna and Linos (2022) do for institutional RCTs.

Multiple comparisons. Our main hypothesis—that directional-finding papers are published at higher rates than null-finding papers—is pre-specified and tested in a single primary regression. The cross-journal and sub-period analyses are exploratory. Applying a Bonferroni correction for the nine robustness specifications in Table 7 yields a corrected threshold of $p < 0.00556$. All main results ($p < 0.001$) clear this threshold; only the *Journal of Finance*-only subset ($p \approx 0.28$) does not, which we interpret as a power issue rather than evidence against the main finding.

8. Conclusion

We provide systematic evidence on significance sorting in the finance literature's treatment of government intervention research. Using 200 in-scope papers spanning 2000–2024, classified and direction-coded with an LLM-assisted pipeline, we document a pronounced cross-sectional pattern: papers with any directional finding appear in the top three finance journals at rates of approximately 88%, compared with only 48% for null or inconclusive findings. A binary directional indicator (our most credible specification, which avoids normative direction coding) yields a probit coefficient of 1.231^{***} (s.e. 0.204), corresponding to a 40 percentage-point direction–publication gap. Directional papers are approximately $1.85\times$ more likely to appear in a top-three journal than null papers. The pattern is concentrated in the *Review of Financial Studies* and *Journal of Financial Economics* and is robust to excluding monetary policy papers and re-estimating via a linear probability model. We cannot detect directional asymmetry at this sample size: positive- and negative-finding papers appear at similar rates, though the power to detect moderate asymmetries is limited.

Exploratory sub-period analysis reveals time heterogeneity. During 2008–2015, the negative-finding premium ($\hat{\beta}_2 = 1.307^{***}$) exceeded the positive-finding premium ($\hat{\beta}_1 = 1.148^{**}$), though the gap is smaller than in earlier analyses; in the post-Dodd-Frank period (2016–2024), the two premiums converge to 0.995 and 1.022. These sub-period estimates are based on small samples and should be read as descriptive patterns, not confirmatory tests.

We interpret these results throughout as a cross-sectional association, not a causal filter effect. Our design cannot distinguish editorial selection from author self-selection in submission or specification search. Two practical implications follow, conditional on the mechanism. For researchers and meta- analysts, the published distribution of findings on government intervention in finance may over-represent papers with directional conclusions; structural bias-correction methods could provide more balanced assessments of policy efficacy. For journal editors, the results are consistent with a case for mechanisms—registered reports, null-result tracks, or results-blind review—that could reduce the significance-sorting pattern, to the extent that it reflects demand-side editorial behavior.

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Appendix A. Keyword Taxonomy

Table A1 lists all terms used in Pass 1 of the classification pipeline.

Table A1. In-Scope Classification: Keyword Taxonomy

Category	Keywords
Financial regulation	regulation, regulatory, Dodd-Frank, Basel, Glass-Steagall, capital requirements, leverage ratio, bank capital
Securities law	SEC, disclosure requirement, Reg FD, Sarbanes-Oxley, SOX, insider trading law, short-selling ban, short-sale restriction
Banking policy	FDIC, deposit insurance, lender of last resort, bailout, TARP, stress test, bank resolution, government guarantee
Monetary/fiscal policy	quantitative easing, QE, Fed intervention, government subsidy, fiscal stimulus
Market microstructure rules	circuit breaker, tick size, trading halt, uptick rule, limit-down, limit-up, trading curb
Corporate governance law	board independence requirement, say-on-pay, proxy access, mandatory audit, governance mandate, clawback provision, mandatory disclosure
General intervention signals	government intervention, government policy, public policy, law and finance, legal protection, shareholder rights, creditor rights, investor protection

Ambiguous standalone terms (*regulation*, *regulatory*, *SEC*) require co-occurrence with at least one additional in-scope keyword before the paper is flagged as a candidate.

Appendix B. LLM Prompts

2.1 In-Scope Classification Prompt

> A finance research paper is 'in scope' for a study of publication bias in government intervention research if and only if its PRIMARY research question is the CAUSAL EFFECT of a specific government law, regulation, or public policy on financial markets, firms, or investors. Exclusion criteria: (1) purely theoretical papers with no empirical test of a policy; (2) papers that only DESCRIBE a regulation without estimating its effect; (3) papers whose main topic is private firm or market behaviour, with policy as incidental context; (4) papers studying central bank communication WITHOUT a specific policy instrument. Return JSON with exactly three keys: in_scope (boolean), confidence (''high'', ''medium'', or ''low''), reason (one sentence).

2.2 Direction Coding Prompt

> The following is the abstract of a finance paper that studies the effect of a government law, regulation, or public policy. Classify the MAIN empirical finding as: +1 if the government intervention produced a POSITIVE or BENEFICIAL outcome (e.g., improved market quality, reduced systemic risk, better investor protection); -1 if it produced a NEGATIVE or HARMFUL outcome (e.g., reduced liquidity, higher costs, unintended consequences, welfare losses); 0 if the finding is NULL, MIXED, or INCONCLUSIVE. Coding rule: code the sign of the estimated coefficient on the primary intervention variable, NOT the authors' normative framing. Return JSON with: direction (1, -1,

```
or 0), confidence (‘‘high’’, ‘‘medium’’, or ‘‘low’’), rationale
(one sentence).
```

Appendix C. Additional Tables

Table A2. Probit Results by Sub-Period *(Exploratory)*

	2000–2007	2008–2015	2016–2024
Positive (+1)	1.600**	1.148**	0.995**
	(0.628)	(0.487)	(0.387)
	[0.37, 2.83]	[0.19, 2.10]	[0.24, 1.75]
Negative (−1)	0.319	1.307***	1.022***
	(0.942)	(0.397)	(0.352)
	[−1.53, 2.17]	[0.53, 2.09]	[0.33, 1.71]
N	28	69	103
Pseudo R^2	0.179	0.163	0.138

Notes. **This analysis is exploratory.** Sub-period sample sizes are small ($N = 28$, 69, 103) and results should be interpreted as descriptive evidence, not confirmatory tests; confidence intervals overlap substantially across periods. Each column estimates equation (1) on the indicated sub-period without additional controls. Standard errors in parentheses; 95% confidence intervals in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Period 2008–2015 spans the financial crisis and post-crisis regulatory response (Dodd-Frank enacted July 2010); 2016–2024 covers the post-implementation period. The apparent asymmetry in the 2008–2015 period (negative > positive) is consistent with but does not causally identify elevated publication rates for research on the costs of the post-crisis regulatory expansion.

Appendix D. LLM Direction Coding: Replication Confusion Matrix

Table A3. LLM Direction Coding: Replication Confusion Matrix

	Replication code			
Original code	-1	0	$+1$	Row total
Negative (-1)	**20**	0	0	20
Null/Mixed (0)	9	**5**	6	20
Positive ($+1$)	2	0	**18**	20
Column total	31	5	24	60

Notes. Entries are paper counts from a 60-paper stratified random subsample (5 papers per source $\times$ direction cell, balanced across the three journals and NBER). Rows show the original direction code used in the main analysis; columns show the code from an independent second LLM run (claude-haiku-4-5-20251001, temperature $= 0$). Bold entries on the main diagonal indicate agreement. Overall agreement: $43/60 = 71.7\%$. Linear-weighted Cohen's $\kappa = 0.674$ ("substantial agreement" by the Landis and Koch (1977) scale). The negative category achieves 100% agreement; the positive category 90%. Disagreements are concentrated in the null/mixed category (25% agreement), with 15 of 20 null-coded papers re-classified as directional in replication—consistent with null abstracts being less decisive and more sensitive to prompt context. Because coding instability pushes ambiguous papers *toward* directional codes rather than toward null, any resulting measurement error attenuates our estimated significance premium.

Appendix E. Data and Replication

All data and code necessary to reproduce the results of this paper are available at `https://github.com/[REPOSITORY]`. The repository contains:

- `requirements.txt` — Python package versions used
- `scripts/01_collect_nber.py` — NBER API data collection
- `scripts/02_collect_journals.py` — Journal data collection
- `scripts/02b_enrich_jfe_abstracts.py` — Elsevier API abstract enrichment for *Journal of Financial Economics*

- `scripts/03_classify_inscope.py` — Two-pass in-scope classification (keyword + LLM)
- `scripts/04_code_direction.py` — LLM direction coding
- `scripts/05_link_nber_to_published.py` — Fuzzy title matching for NBER-to-journal linkage (Round 1)
- `scripts/05b_improve_matching.py` — API-based abstract matching for unmatched NBER papers (Round 2)
- `scripts/05c_fulltext_matching.py` — LLM full-text matching via NBER PDFs (Round 3)
- `scripts/05d_published_fulltext_matching.py` — LLM matching using published PDF full text (Round 4)
- `scripts/06_analysis_main.py` — Primary probit regressions and summary statistics
- `scripts/08_robustness.py` — Robustness checks
- `scripts/09_code_id_strategy.py` — LLM identification-strategy classification (RDD, IV, DiD, Event Study)
- `scripts/10_collect_afa.py` — AFA annual meeting program collection (2006–2024)
- `scripts/10b_enrich_afa_abstracts.py` — Abstract enrichment for AFA papers via OpenAlex, CrossRef, and NBER
- `scripts/10c_fix_2014_2016_parsing.py` — Corrected HTML parsing for 2014–2016 AFA program pages
- `scripts/11_afa_analysis.py` — AFA-baseline probit regressions (AFA conference baseline, Section 6)
- `scripts/11c_recode_afa_direction_llm.py` — LLM direction re-coding for AFA papers (uniform coding robustness check)
- `scripts/04b_code_direction_truncated.py` — Direction coding from first 100 words only (symmetric-window robustness check)
- `data/` — Final analysis dataset in Parquet format

- `validation/human_coding_sheet_40.xlsx` — Coding sheet for 40-paper human direction-coding validation (to be filled by authors)
- `scripts/13_human_validation_kappa.py` — Computes Cohen's κ between human and LLM codes once coding sheet is complete
- `paper/` — LaTeX source for this paper

Appendix F. In-Scope Classifier Validation

Direction-coding reliability. The primary validation evidence for our LLM pipeline concerns direction coding, reported in the main text (Section 3.4). We re-ran the direction-coding protocol on a stratified random sample of 60 papers and treated the second run as an independent rater. Overall linear-weighted Cohen's $\kappa = 0.674$ ("substantial agreement"); agreement is near-perfect for directional papers (-1: 100%, $+1$: 90%) and concentrated in the null/mixed category (25%). The confusion matrix is reported in Appendix Table A3.

In-scope classifier design. The in-scope LLM classifier is applied to all keyword-flagged papers after a broad keyword filter selects candidates from the raw journal and NBER corpora. The keyword list is deliberately over-inclusive (any mention of regulation, law, intervention, or policy in the title or abstract), so the classifier's primary role is removing false positives from the keyword stage rather than detecting in-scope papers from scratch.

A conservative bound on the in-scope false-positive rate is available from the data itself: all 200 in-scope papers were independently verified as on-topic during the matching and scope-review stages of the pipeline (Sections 3.3 and 3.5). No paper was flagged in-scope by the LLM and subsequently removed as out-of-scope during these downstream checks, giving a post-hoc precision of 100% within the reviewed portion of the sample.

The more material concern is the false-negative rate—papers that are genuinely in-scope but missed by the keyword filter or misclassified as out-of-scope by the LLM. False negatives would reduce the representativeness of our sample but would not mechanically bias the direction-publication association unless missed papers systematically differ in direction from included papers. Given that the keyword filter captures all standard regulatory and supervisory terminology in finance, we expect any false negatives to be idiosyncratically missed rather than directionally selected.

Appendix G. Human Direction-Coding Validation

Table A4. Human vs. LLM Direction-Coding Agreement (N = 40)

	LLM code			
Human code	−1	0	+1	Total
−1	*[–]*	*[–]*	*[–]*	*[–]*
0	*[–]*	*[–]*	*[–]*	*[–]*
+1	*[–]*	*[–]*	*[–]*	*[–]*
Total	*[–]*	*[–]*	*[–]*	40

Notes. Stratified random sample of 40 papers (20 published, 20 NBER). Each paper was coded by the authors using only the abstract, without reference to the LLM output. Linear-weighted Cohen's κ (direction, 3-category): *[–]*. Cohen's κ (binary directional): *[–]*. Results to be completed before final submission. Human coding sheet: `validation/human_coding_sheet_40.xlsx`. Script to compute κ: `scripts/13_human_validation_kappa.py`.